

Executive Summary & Domain Overview:

This report provides a comprehensive summary of all engineering tasks completed across **Lab 1** (Requirements Engineering, UML Use-Case Modeling, and Specification Documentation) and **Lab 2** (Agile Backlog Creation, Jira Kanban, Scrum Sprint Simulation, Burndown Analytics, and Bug Tracking). The scenario addresses *Problem Statement #42*: an enterprise-grade developer portal that maintains a centralized searchable microservice catalog, renders interactive dependency graphs, executes periodic 30s health checks, and triggers real-time multi-channel incident alerts.

1. Lab 1: Requirements Engineering & Modeling Achievements

Req ID	Type / Domain	Requirement Description	Priority	Status
FR-001	Health Monitoring	Ping microservice health endpoints at 30s intervals; compute rolling 24-hr availability percentage.	High	Completed
FR-002	Catalog Management	Searchable registry storing metadata: name, owner team, version, repo URL, environments, and API docs.	High	Completed
FR-003	Dependency Mapping	Auto-discover and render interactive upstream/downstream dependency graph with node drill-down.	High	Completed
FR-004	Alerting & Notifications	Dispatch real-time downtime alerts via Email, Slack, and Webhook on 3 consecutive failed probes.	Medium	Completed
FR-005	API Documentation	Aggregate and render auto-generated OpenAPI / Swagger specs with interactive payload schemas.	Medium	Completed
NFR-001	Performance & Scalability	Dependency graph renders up to 200 nodes within 2.0s and interaction latency ≤ 200ms under 50 peak users.	High	Completed
NFR-002	Security & Compliance	All probes encrypted via TLS 1.2+, Enterprise SSO authentication (OAuth 2.0/SAML), encrypted secrets vault.	High	Completed

Lab 1 Diagrammatic & Specification Deliverables:

- Primary UML Use-Case Diagram:** Fully compliant diagram built in StarUML (.mdj) and draw.io (.drawio) capturing Actors (DevOps Engineer, System Architect, On-Call Engineer, Enterprise IdP, Monitored Microservice), System Boundary, and standard «include» and «extend» relationships.
- Alternate Flow UML Diagram:** Dedicated lifecycle modeling outage detection, threshold evaluation (3 failed probes), multi-channel alert dispatch, and auto-recovery notifications.
- Use-Case Flow Specification:** 1-page formal document detailing Preconditions, Postconditions, Main Success Scenario, and Alternate Flows for *Monitor Service Health*.
- Exception Flow Specification:** 1-page fault handling guide detailing recovery routines for Network Timeouts (EF-001), Notification Delivery Failures (EF-002), TLS Violations (EF-003), Vault Outages (EF-004), and SSO Token Expiry (EF-005).

2. Lab 2: Agile Jira Simulation & Project Execution

A. Kanban Project (Kanban_BPS#42 / KBPS42):

- Configured Company-Managed Kanban board with stages: *TO DO* → *IN PROGRESS* → *IN REVIEW* → *DONE*.
- Added all 7 requirements (5 FRs + 2 NFRs) and structured 7 corresponding Epics linked via parent relationships.
- Authored user stories ("As a... I want... So that...") and added engineering subtasks (e.g., HTTPS pinger engine, metadata validation forms, Slack webhook dispatchers).
- Generated **Kanban.PDF** with 6 sequential screenshots and detailed technical descriptions.

B. Scrum Project (Scrum_BPS#42 / SBP42):

- Configured Scrum Software Development project with product backlog and sprint timeboxing.
- Estimated all backlog items using the **Fibonacci sequence** (1, 2, 3, 5, 8) totaling **29 Story Points**.
- Simulated **Sprint 1** (20 SP) and **Sprint 2** (9 SP) with task transitions across To Do → In Progress → Done.
- Generated official Jira **Burndown Chart** validating linear velocity and work completion against the ideal guideline.
- Answered all 4 Agile post-sprint reflection questions in **Scrum.PDF**.

C. Bug Tracking Project (BugTracker_BPS#42 / BTBPS42):

- Created dedicated defect tracking space; logged 4 categorized defects across health checks, catalog uniqueness, graph performance, and webhook encoding.
- Documented root cause analysis and engineering remediation strategies in **BugReport.PDF**.

3. Complete Deliverables Index & Repository Status

Lab	Deliverable Name	Format & File Reference	Status / Submission Target
Lab 1	Requirements Table & Specs	requirements.pdf, use_case_flow_specification.pdf	Verified & Pushed to GitHub
Lab 1	UML Use-Case Diagrams	use_case_diagram.drawio/.mdj/.png	Verified & Pushed to GitHub
Lab 2	Kanban Project Report	Kanban.PDF (6 Screenshots)	Ready for Google Form Upload
Lab 2	Scrum Project Report	Scrum.PDF (Burndown Chart + Reflections)	Ready for Google Form Upload
Lab 2	Bug Tracking Report	BugReport.PDF (Defect Inventory & RCA)	Ready for Google Form Upload
Lab 2	Merged Master Report	Subramani_BM_PES1UG24CS473-BPS#42.pdf (8 Pages)	Combined 3-in-1 PDF